

AI Engineer

Step 1: Learn Python & Mathematics

CareerCompass • Detailed Concept Notes • Research-backed learning guide

Concept & Learning Objective

AI engineering requires both software and quantitative foundations. Python provides the ecosystem; mathematics provides the language for understanding data, optimization and model behavior.

Focus on the parts of mathematics that repeatedly appear in ML: vectors/matrices, probability, statistics, derivatives/gradients and optimization. The goal is intuition plus enough formal understanding to interpret model equations and debugging signals.

Core Concepts — Detailed Breakdown

- Python: functions, classes, modules, exceptions, virtual environments and package management.
- NumPy: arrays, shapes, broadcasting, vectorized operations and matrix multiplication.
- Linear algebra: vectors, matrices, dot products, norms and transformations.
- Statistics: mean, median, variance, standard deviation, covariance and correlation.
- Probability: random variables, conditional probability, distributions and expectation.
- Calculus: derivatives and partial derivatives as rates of change.
- Optimization: loss functions, gradients, learning rate and local search intuition.
- Numerical thinking: floating-point precision, normalization and data scale.

What You Should Be Able To Do

- Implement vector and matrix operations with NumPy.
- Calculate mean/variance and interpret them.
- Explain conditional probability.
- Explain why a gradient can guide optimization.
- Use a virtual environment and requirements file.

Hands-on Project / Practice

- Implement linear regression with gradient descent using NumPy. Plot training loss over iterations and experiment with learning rate.

Recommended References

- Python Documentation — <https://docs.python.org/3/>
- TensorFlow Learn — <https://www.tensorflow.org/learn>
- Scikit-learn User Guide — https://scikit-learn.org/stable/user_guide.html

Note: These notes are designed to teach the concepts and provide a practical learning path. Always prefer the linked official documentation for current APIs, SDK versions and platform-specific release requirements.